

Cryptocurrency Momentum, Reversal, Funding, Open Interest, and Liquidation Cascades

Executive summary

The strongest evidence in liquid crypto points to **short-horizon continuation, not blanket mean reversion**. Across BTC, ETH, and the more liquid altcoins, peer-reviewed and credible working-paper evidence finds momentum at intraday, daily, and weekly horizons, especially when the asset universe is filtered toward larger and more tradable coins. By contrast, the cross-sectional **short-term reversal effect is concentrated in small, illiquid, low-volume coins**; once liquidity screens are imposed, reversal weakens sharply and often flips into momentum for larger coins. This is the single most important result for anyone trying to trade “Qullamaggie-style” continuation patterns in liquid crypto rather than paper portfolios of microcaps. ¹

Funding rates and open interest are useful, but mostly as **state variables** rather than magic directional predictors. Academic and institutional work is fairly consistent on three points: perpetual-futures prices matter for price discovery; funding carries real information about crowding, convenience yield, and cash-and-carry profitability; and open interest becomes much more informative when read jointly with price. What is much weaker is the claim that raw funding alone reliably predicts the next move in a single asset. Practitioner tests find limited single-asset forecasting power, while academic work shows stronger value in carry/arbitrage frameworks and in explaining futures-spot wedges. ²

For liquidation cascades, the most defensible conclusion is conditional: **forced deleveraging can create tradable reversal zones, but only after leverage has actually been cleared**. Practitioner work and newer working papers show that liquidation clusters, open-interest collapses, and one-sided positioning often mark local bottoms or tops after squeezes. But the same literature also shows that thin liquidity can allow the move to overshoot before stabilizing. In other words, “buy the puke” or “fade the vertical move” works best *after* evidence of position closure and liquidity re-absorption, not during the first impulse. ³

For the specific pattern family you asked about, the balance of evidence is this. **Continuation breakouts** in liquid crypto are the best supported. **Episodic pivots** have partial support, mainly through abnormal-return continuation, jump/news evidence, and volume/trend signals, but the evidence is materially weaker than in equities because crypto lacks a standardized earnings cycle. **Parabolic reversals** are real, but the edge is horizon- and regime-dependent: extreme moves often continue first, then reverse only once leverage and positioning unwind. ⁴

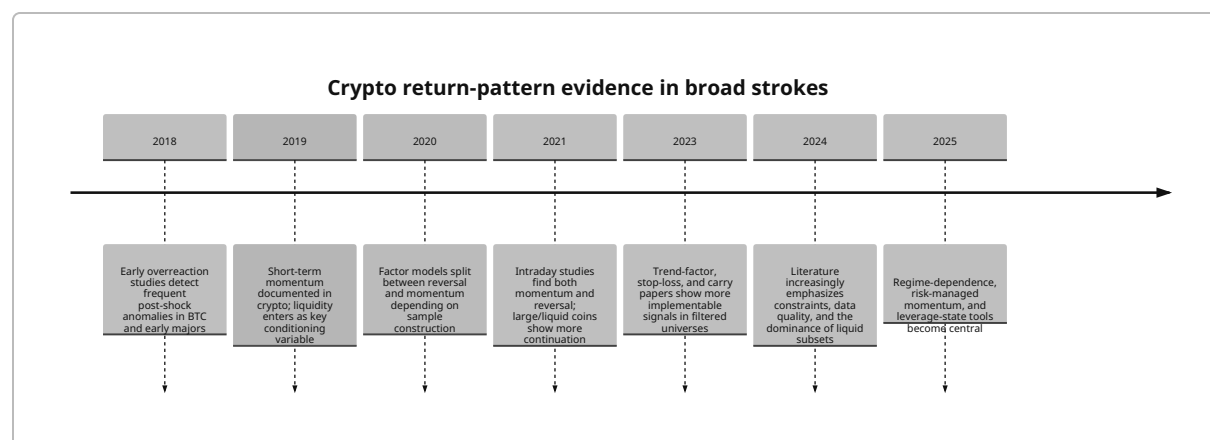
Evidence base and how to read it

This literature is unusually sensitive to **sample design**. The biggest split in findings comes from whether studies include thousands of tiny, low-volume coins or restrict the sample to large, liquid names. That matters because several anomalies that look large in broad cross-sections are driven by microcaps of

limited economic importance, while momentum-type effects are more likely to persist in large-cap and liquid subsets. Recent survey-style work using broad CoinGecko panels tries to mitigate survivorship bias by including inactive coins, but also explicitly warns about wash trading, unreliable early exchange data, and the need to emphasize post-2020 and reputable-venue samples. ⁵

Methodologically, the evidence here comes from several distinct traditions. Asset-pricing papers use cross-sectional sorts, factor models, and Fama-MacBeth regressions on weekly or monthly returns. Market-microstructure papers use high-frequency BTC or multi-exchange order-book data. Perpetual-futures papers split into no-arbitrage theory, basis/funding empirics, and event studies around stress episodes. Practitioner research from Coinbase Institutional, Glassnode, Kaiko, and Presto is usually strongest on derivatives positioning, liquidation mechanics, and implementation reality, though it is less standardized than peer-reviewed work. ⁶

The timeline below summarizes the broad arc of findings that recur across the highest-credibility sources. It is a synthesis of the cited papers and reports, not a claim that the literature moved in neat discrete blocks. ⁷



Momentum and reversal in crypto

The broad academic case for crypto momentum is not controversial anymore; the controversy is **where** it survives and **at what horizon**. Liu and Tsyvinski show that cryptocurrency returns are predictable from crypto-specific factors and that momentum is one of the economically important regularities in the asset class. Later work sharpened the result by horizon. Tzouvanas, Kizys, and Tsend-Ayush find that momentum is **highly significant for short-term portfolios** and fades at longer horizons. Borgards finds that dynamic time-series momentum periods are larger and longer in cryptocurrencies than in U.S. stocks, with crypto momentum strategies outperforming buy-and-hold on a risk-adjusted basis. Filippou, Rapach, and Thimsen also find that momentum, size, and value-type predictors remain among the most important inputs in out-of-sample machine-learning forecasts. ⁸

The most important nuance is liquidity segmentation. Begušić and Kostanjčar find a **strong momentum effect in the most liquid cryptocurrencies**, and propose long-only tilts toward liquid winners and illiquid losers. Zaremba and coauthors then show why earlier papers disagreed with one another: the “reversal” seen in many broad samples is largely an artifact of illiquid names. In their peer-reviewed study, daily reversals arise from the illiquidity of the vast majority of traded coins, while the handful of largest and most

tradeable coins exhibit **daily momentum rather than reversal**. Milan Fičura's working paper pushes this further: in large, liquid coins, short-term momentum is strongest when the signal is defined relative to prior highs, while small and illiquid coins carry the reversal effect. ⁹

That prior-high result matters for breakout trading. Fičura finds that “high-momentum,” measured as distance from the previous multi-week high, is a **superior predictor** of future returns in crypto, with the one-week version predicting positively for large, liquid coins and negatively for small, illiquid ones. A more recent behavioral extension outside your core window, Jia's 2026 anchoring paper, also reports a positive “nearness to 52-week high” return spread, which reinforces the same broad idea: in crypto, being near a prior high is often a continuation feature rather than an automatic fade signal. Even staying within 2016–2025, the weight of evidence points in that direction for liquid names. ¹⁰

Intraday evidence is similarly mixed in sign but clear in structure. Wen, Bouri, Xu, and Zhao document both **intraday momentum and reversal** in Bitcoin, with related evidence in ETH, LTC, and XRP, and show that the pattern changes around intraday jumps, FOMC announcements, liquidity conditions, and COVID. Caporale and Plastun's overreaction studies add another layer: after one-day abnormal returns, prices often continue in the same direction through the day of the move and sometimes into the following day, but simple exploitable profits are far less robust once trading implementation is considered. Borgards and Czudaj find that price overreactions are highly prevalent across frequencies, again suggesting that the jump from anomaly to tradeable alpha is not automatic. ¹¹

More recent cross-sectional work suggests that trend extraction can be made more robust. Fieberg and coauthors' CTREND factor aggregates price and volume information across horizons and remains robust across subperiods, market states, and alternative research designs, including transaction costs and large/liquid subsamples. Tan and Tao, using nearly 3,000 coins from 2013 to 2022, find that **price-based technical signals are more effective at short horizons**, while **volume-based signals are more useful at longer horizons**. Hsieh, Huang, and Liu then show that momentum profits in crypto are strongest in **UP–UP regime transitions**: if the market is already in an up state and stays there, continuation survives; other transitions suppress it. ¹²

Short-term reversal, then, is not absent. It is just **poorly generalized** if you trade liquid crypto. Shen, Urquhart, and Wang's three-factor model uses reversal as the third factor over 2013–2019, but Jia, Goodell, and Shen later argue that a momentum factor fits the more recent sample better. The literature has effectively moved from “crypto has reversal” to “crypto has reversal in some segments and momentum in others, with sample period and tradability doing most of the work.” ¹³

Funding, carry, and open interest

Perpetual futures matter because they are not just a derivative overlay; they are often where **price discovery** happens. Alexander and Heck show that BTC perpetual swaps and futures on large offshore venues were the strongest instruments for Bitcoin price discovery, while regulated CME futures and U.S. spot markets were more reactive than leading during their sample. Cosenza's later work similarly identifies Binance spot and perpetual futures as primary sources of Bitcoin price discovery, with lower transaction costs and higher trading volumes enhancing that role. For signal design, this means perp data should not be treated as mere confirmation of spot; in many episodes, it is the leader. ¹⁴

Funding itself is structurally important. Coinbase's primer notes that a displayed 0.01% funding rate can annualize very differently depending on interval length, and that the common use of clamping functions and base rates creates a **positive skew** to baseline funding, often incentivizing short-perp/cash-and-carry arbitrage. Akerer, Hugonnier, and Jermann, along with He, Manela, Ross, and von Wachter, provide the formal no-arbitrage framework: perpetual prices need not equal spot at all times; instead, they are pinned within bounds that depend on funding design and frictions. Hugonnier and Jermann's 2025 extension shows that clamp design itself can create the bimodal distribution of futures-to-spot ratios observed at higher frequencies in BTC and ETH. ¹⁵

Empirically, the most robust directional result is not "funding predicts next return" in a simple way. It is closer to "funding and basis summarize crowding, trend chasing, and arbitrage scarcity." Schmeling, Schrimpf, and Todorov show that crypto carry can become very large, sometimes above 40–60% annualized, and interpret it as the outcome of **trend-chasing leveraged demand plus scarce arbitrage capital**. He et al. find that past returns significantly explain the futures-spot gap: when past returns are high, perpetual futures trade richer relative to spot, which they interpret as **positive-feedback or momentum trading behavior** in perpetual markets. In other words, crowded positive funding often confirms a continuation regime before it becomes a contrarian signal. ¹⁶

On direct predictability, the evidence is narrower. Presto's single-asset test on Binance data finds that 7-day funding-rate changes explain some *contemporaneous* 7-day price variation, but nearly zero out-of-sample power for predicting the **next** interval's price change in a single asset. Their own conclusion is that funding is likely more useful **cross-sectionally across many assets** than as a one-coin directional trigger. A late-2025 SSRN paper by Emre Inan adds that funding rates themselves are forecastable one step ahead for BTC on Binance and Bybit, so traders can model future carry, but that is not the same as predicting the sign of the next spot move. ¹⁷

The best way to think about funding for implementation is as a **carry-and-crowding variable**. Borri, Liu, Tsyvinski, and Wu report that the profitability of the short-perp/long-spot crypto carry strategy from 2020 to 2025 was mostly driven by the funding leg, with about 8% mean return and low volatility in their sample, though they also note that profitability compressed sharply after 2024 and turned negative in 2025. That is a useful warning: funding-harvest strategies can look extraordinarily good until market structure changes and the premium compresses. ¹⁸

Open interest is best treated as a *joint* signal with price, not a stand-alone one. CME's generic futures guidance says rising open interest usually confirms trend strength and falling open interest can indicate weakening conviction. In crypto, Glassnode's 2025 LPOC framework operationalizes that idea by classifying the joint movement of price and OI into long openings, short openings, long closures, and short closures. Their BTC and ETH case studies claim that **extreme long openings often precede tops**, while **long-closure spikes often mark local bottoms**, especially in sustained uptrends and after stronger deleveraging events. This is not peer-reviewed academic evidence, but it is one of the clearest institutional signal frameworks built on exchange-level derivatives data. ¹⁹

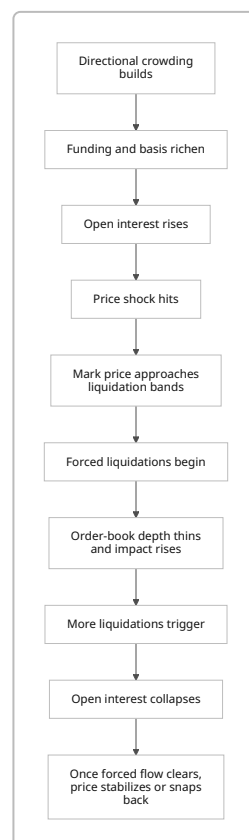
There is also evidence that open interest helps explain broader derivatives states. Schmeling, Schrimpf, and Todorov report that OI enters positively in regressions for basis, consistent with trend-chasing and price pressure. Coinbase and Kaiko both use OI-plus-funding states to interpret whether a rally is overheating or whether a flush has likely run its course: for example, Kaiko described BTC's March 2024 rally as accompanied by a large OI increase and still-high funding, while Coinbase's August 2023 post-cascade note

argued that the drop in OI together with normalization in OI-weighted funding reduced the probability of another large liquidation wave. These are useful as conditional risk tools, but they should not be confused with a universal forecasting law. ²⁰

Liquidation cascades and reversal mechanics

Liquidation cascades are the point where crypto market microstructure departs most sharply from textbook factor investing. The basic mechanism is simple: leveraged positions are marked to a mark price; if equity falls below maintenance requirements, the venue forcibly closes the position; that flow hits the order book, moves price further, and can trigger the next layer of liquidations. Binance's documentation explains the role of mark price and maintenance margin in preventing unnecessary liquidations while still closing under-margined positions. Glassnode's liquidation-heatmap research shows the same idea in a trading framework: leverage concentrates risk around price bands, and once price enters those bands, forced liquidations can amplify volatility. ²¹

The schematic below summarizes the way the cited practitioner and academic sources describe a typical crypto liquidation cascade. It is a synthesis of exchange mechanics, Glassnode's heatmap work, and newer working papers on runs in perpetual futures. ²²



Direct empirical work on crypto liquidation episodes is still thin relative to momentum papers, but the evidence that exists is increasingly consistent. Glassnode's LPOC work argues that **long-closure surges often mark local bottoms** and that stronger deleveraging events increase the likelihood of a rebound. Its liquidation-heatmap report finds that liquidation clusters align with subsequent liquidation spikes and can

act as “structural pressure points.” Coinbase’s institutional commentary after the August 2023 flush argued that a sharp OI decline and normalized funding likely limited the scope for further cascading liquidations. A January 2026 Coinbase note put the same point more plainly: a liquidation-driven reset can support a bounce, but it is not a substitute for durable demand. ²³

The caution is that bottoms are rarely clean. Glassnode explicitly notes that thin liquidity can leave price vulnerable to **downside continuation**, even after leverage has started to clear. Their 2026 reports describe dense liquidation bands below spot as “liquidity magnets,” where price can be drawn lower before the forced selling is fully absorbed. That matches the logic in Joshi and Chance’s “run on open interest” working paper, which frames cascades as a coordination problem in which higher adverse-selection costs can trigger exits from open futures positions. ²⁴

Event evidence from 2025–2026 develops the same picture at finer resolution. A minute-level working paper on the October 2025 crash states that futures led the crash and that mark price undershot both spot and perp trade prices during the event. A linked paper on Hyperliquid’s post-shock recovery describes a **two-regime** response: inside-market pricing recovers within weeks, while quoted depth recovers more slowly. That is exactly the environment in which a reversal can be directionally right but still hard to monetize at size because liquidity remains impaired. ²⁵

So the best-supported reversal mechanism in liquid crypto is not generic “oversold bounce.” It is **post-liquidation rebound after one-sided leverage is extinguished**. That distinction matters. A trader fading a parabolic move before OI has broken and liquidation pressure has passed is stepping in front of forced flow. A trader waiting for OI collapse, funding normalization, and evidence of passive bids returning is trading after the forced-flow regime has already changed. The literature and practitioner evidence favor the second approach. ²⁶

What the evidence says about Qullamaggie-style patterns

There is no academic paper that tests “Qullamaggie continuation breakout,” “episodic pivot,” or “parabolic reversal” under those labels. The evaluation here is therefore an inference from the cited literature. I am mapping those trading patterns onto empirically testable families: breakouts through prior highs with volume/trend confirmation, catalyst-driven gap-and-go continuation, and exhaustion reversals after vertical extensions. The strength of the conclusion differs across the three. ²⁷

For **continuation breakouts**, the evidence is favorable in liquid crypto. Large/liquid coins show momentum where smaller coins show reversal. High-momentum relative to prior highs predicts future returns in liquid names. Trend factors built from price and volume remain robust after transaction costs and in liquid subsets. Momentum is also strongest when the broader market stays in an up regime, which lines up with the trader intuition that the best breakouts occur when the tape is already in expansion mode. If you translate the pattern into data language, the most defensible breakout setup is: liquid asset, prior-high proximity, positive trend factor, rising but not yet euphoric OI, and a background UP–UP regime. ²⁸

For **episodic pivots**, the evidence is positive but weaker and less direct. Caporale and Plastun find continuation after one-day abnormal returns, at least over the day of the shock and sometimes into the next day. Wen et al. find that intraday momentum and reversal depend on jump conditions and liquidity, while Cepni, Aysan, and Caporin show that information releases raise the probability of intraday price jumps. Tan and Tao’s result that price-based signals dominate at short horizons and volume-based signals

matter at longer horizons also fits the intuition of a catalyst-driven repricing that then broadens into participation. What is missing, compared with equities, is a standardized catalyst taxonomy like earnings surprises or guidance revisions. Crypto has ETF approvals, listings, unlocks, or protocol events, but the literature is much less organized around them. ²⁹

For **parabolic reversals**, the evidence is genuinely mixed and highly horizon-dependent. One branch of the literature says overreactions are common and reversals frequent across intraday and interday windows. Another says that coins with the strongest extreme positive returns — the MAX effect — often continue to outperform, not mean-revert, in the following period. Grobys and coauthors then add a warning from the risk side: crypto momentum can suffer violent crashes from single-name blowups in the short leg, and these crashes are **not** necessarily classic market-wide reversals. The practical implication is that fading vertical price action too early is dangerous. The better-supported reversal trade is specifically the **post-squeeze, post-liquidation** reversal, not the first fade attempt against strength or weakness. ³⁰

A useful way to rank the three patterns is by how directly the literature supports them in BTC, ETH, and liquid alts. Continuation breakouts rank first. Episodic pivots rank second, with more inference required. Parabolic reversals rank third unless they are explicitly tied to leverage exhaustion, OI collapse, and liquidation clearing. ³¹

Implementation, risk, and paper comparison

The main implementation lesson is brutal but simple: **many crypto anomalies look best where you least want to trade them**. Fieberg, Liedtke, and Zaremba show that size and volume anomalies originate in micro-cap coins of negligible economic importance, while momentum survives in larger cryptocurrencies. Cakici and coauthors find that many machine-learning alphas remain profitable after trading costs, but the alpha is concentrated in hard-to-trade, small, illiquid, volatile coins. Fičura shows the same split for reversal versus momentum: short-term reversal is driven mostly by low-volume coins that may be difficult to buy at the reported close, while the momentum effect is more robust in larger-cap liquid coins. ³²

Tail risk is not an afterthought here. Grobys and coauthors find that equal-weighted large-cap momentum can still crash severely; in their sample, a single short-leg coin drove a **-255% monthly crash** in December 2020, and volatility management improved risk-adjusted returns without changing the tail-risk character of the strategy. By contrast, Yang's 2025 risk-managed momentum paper reports improved average weekly returns and Sharpe ratios relative to conventional momentum, and argues that crypto does not show the same extended "momentum crash" pattern familiar from equities. Both can be true: volatility targeting can improve average outcomes, while left tail and right-tail jump exposure remain nasty in discrete events. ³³

Transaction costs and market impact differ sharply by venue and strategy. In He et al.'s BTC/ETH perpetual arbitrage setting, direct fees dominate indirect costs; using Kaiko order-book snapshots, they estimate less than 0.05 bps average perp spread and less than 0.3 bps impact for a \$0.5 million trade, far below the fee assumptions they model. But that result belongs to **arbitrage in deep BTC/ETH perps**, not to momentum-chasing in thinner altcoins. Ruan and Streltsov also show that spot market quality worsens around perpetual funding-settlement windows because informed trading rises and market makers widen spreads. If you implement momentum or breakout strategies around funding events, you should assume adverse selection is higher exactly when crowding is largest. ³⁴

Carry trades have their own path risk. Schmeling, Schrimpf, and Todorov show that crypto carry is attractive on average but fundamentally a story of trend-chasing demand and scarce arbitrage capital. Their futures-leg path analysis highlights bankruptcy risk under leverage: the chance of the return before maturity falling below the leverage-implied bankruptcy threshold is very high at common leverage levels. That means even “delta-neutral” or basis-style strategies can be operationally fragile if financed aggressively. ³⁵

Stop-loss and state filters are therefore not cosmetic. Sadaqat and Butt find that stop-loss-enhanced momentum across 147 cryptocurrencies from 2015 to mid-2022 improves returns, Sharpe ratios, and alphas relative to benchmark momentum strategies, while Hsieh, Huang, and Liu show that regime continuity matters for whether momentum works at all. For implementation in liquid crypto, that argues for a layered decision rule: momentum signal first, then liquidity filter, then regime filter, then crowding filter from funding/OI, then explicit exit discipline. ³⁶

Concise comparison of key sources

Source	Data and method	Main empirical result	Relevance to Qullamaggie-style patterns
Liu & Tsyvinski 2021, <i>RFS</i> ³⁷	Major cryptocurrencies; asset-pricing tests	Crypto returns are predictable from crypto-specific factors, including momentum-type effects.	Supports the idea that persistent trend signals exist in crypto at all.
Tzouvanas, Kizys & Tsend-Ayush 2020, <i>Economics Letters</i> ³⁸	Weekly momentum portfolios	Momentum is highly significant for short-term portfolios but fades at longer horizons.	Direct support for breakout continuation over short horizons, not for long hold times.
Zaremba et al. 2021, <i>IRFA</i> ³⁹	Broad cross-section; liquidity conditioning	Daily reversal is mostly an illiquidity effect; largest, most tradeable coins show daily momentum.	Strong evidence for continuation in liquid crypto and against relying on reversal in majors.
Wen et al. 2022, SSRN / later journal version ⁴⁰	High-frequency BTC, plus ETH/LTC/XRP checks	Both intraday momentum and reversal exist; patterns vary with jumps, FOMC, liquidity, COVID.	Supports episodic pivots and intraday continuation, but warns that regime matters.
Fičura 2023 working paper ⁴¹	Size- and volume-filtered cross-section	Prior-high “high-momentum” predicts positively for large/liquid coins and negatively for small/illiquid coins.	Probably the closest academic evidence to liquid-crypto breakout logic.

Source	Data and method	Main empirical result	Relevance to Qullamaggie-style patterns
Fieberg et al. 2025, trend factor ⁴²	3,000+ coins; ML-based trend aggregation	CTREND predicts cross-sectional returns and survives transaction costs, including in large/liquid coins.	Strong support for systematic continuation signals beyond simple past-return sorts.
Hsieh, Huang & Liu 2025, <i>FRL</i> ⁴³	Regime-transition conditioning	Crypto momentum is strongest only in UP-UP states; other transitions suppress it.	Very relevant for selecting which breakout environments to trade.
Schmeling, Schrimpf & Todorov 2023, BIS/SSRN ³⁵	BTC/ETH futures basis across exchanges	Carry can be extremely large; driven by trend-chasing demand and scarce arbitrage capital; leverage risk is high.	Funding/basis extreme is informative, but also a warning that crowding can persist before snapping.
He et al. 2024, <i>Fundamentals of Perpetual Futures</i> ⁴⁴	No-arbitrage theory plus BTC/ETH and other perp data	Past returns positively explain perp-spot wedges; profitable arbitrage survives fees in major perps.	Funding and perp premium often confirm momentum rather than immediately reversing it.
Glassnode 2025 LPOC and liquidation heatmaps ⁴⁵	Price + OI signal design; liquidation mapping	Extreme long openings often precede tops; long closures often mark local bottoms; liquidation bands flag pressure points.	Most relevant institutional framework for timing parabolic reversals <i>after</i> deleveraging.
Presto Research 2024 ⁴⁶	Binance funding-rate tests	Funding has weak next-interval single-asset predictive power, but may work cross-sectionally.	Good reason not to use funding alone as a directional trigger.
Grobys et al. 2025, <i>Financial Markets and Portfolio Management</i> ⁴⁷	Top-30 weekly momentum, 2016–2023	Momentum crashes can be severe and idiosyncratic; volatility management helps but does not remove tail risk.	Strong warning against concentrated short legs and naïve parabolic fades.

Open questions and bottom line

The literature still has gaps. Direct peer-reviewed evidence on **open interest as a stand-alone return predictor** is much thinner than the evidence on momentum or carry, and most actionable OI work still

comes from institutional analytics rather than journals. The same is true for **liquidation-cascade reversals**: the mechanism is widely accepted, but the sharpest direct evidence is in 2025–2026 working papers and practitioner studies, not yet in a deep peer-reviewed canon. Venue coverage also matters, because liquidation, funding, and OI series are constructed differently across exchanges and vendors. ⁴⁸

The bottom line is sharper than the literature's surface disagreement suggests. If the target is **BTC, ETH, SOL, and other liquid crypto assets**, the evidence tilts toward **continuation** over short to intermediate horizons, especially near prior highs, in persistent up regimes, and when price and OI are rising together without already showing post-euphoria unwind. Evidence for **short-term reversal** is strongest in illiquid names and during forced deleveraging episodes, which makes it far less useful as a default style in liquid majors. **Funding** is best used as a crowding/carry variable, **open interest** as a price-conditioned confirmation or exhaustion gauge, and **liquidation metrics** as the most credible trigger for trading reversal *after* a move has broken itself. ⁴⁹

If you translate that into a practical research prior for Qullamaggie-style crypto trading, it is this: systematic edges in liquid crypto are most likely to come from **breakouts that align with trend, liquidity, and derivatives confirmation**, not from fading every vertical move. The reversal trade becomes strongest only when the market has already paid the leverage bill. Before that moment, the tape can stay irrational longer than a clean chart pattern can stay intact. ⁵⁰

¹ ²⁸ ³¹ ³⁹ ⁴⁹ Up or down? Short-term reversal, momentum, and liquidity effects in cryptocurrency markets

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- 34 44 Fundamentals of Perpetual Futures We are grateful to Lin William Cong, Urban Jermann, Shimon Kogan, Tim Roughgarden, Adrien Verdelhan, as well as conference participants at the 2024 Utah Winter Finance Conference and seminar participants at a16z Crypto, Hebrew University, Reichman University, and the Virtual Derivatives Workshop for their insightful feedback and helpful comments. Songrun He is at Washington University in St. Louis (h.songrun@wustl.edu). Asaf Manela is at Washington University in St. Louis and Reichman University (amanela@wustl.edu). Omri Ross is at University of Copenhagen (omri@di.ku.dk), and Chief Blockchain Scientist at the eToro Group. Victor von Wachter is at University of Copenhagen (victor.vonwachter@di.ku.dk).
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43 State transitions and momentum effect in cryptocurrency market

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